Employee Satisfaction and Long-run Stock Returns, 1984-2020

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Abstract

Economic theory predicts that (in the absence of mispricing) the excess return to socially responsible businesses is negative in equilibrium. In contrast, using the state-of-art empirical models and a sample spanning four decades (1984-2020), an equal-weighted portfolio of companies that treat their employees the best earns an excess return of 2% to 2.7% per year. The estimated alphas are positive in most periods within the sample (with no upward or downward trend) and are particularly large during crisis periods. Overall, the results suggest that the stock market (still) undervalues employee satisfaction.

Keywords: Employee satisfaction, Business social responsibility, Socially responsible investing, Underreaction, Corporate culture

JEL Classifications: G14, G38, J28, M14

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1. Introduction

The volume of assets under management that incorporates some element of socially responsible investing (SRI) has grown exponentially over the past decade, passing $35 trillion at the start of 2020 or about 40% of all professionally managed assets.[1] These investments are driven by strategies that consider at least one dimension related to environmental, social, and governance (ESG) factors in portfolio selection. Whether or not these investment strategies outperform has been a source of debate among academics and practitioners.

Our focus on this paper is on one aspect of corporate social responsibility: employee satisfaction. Theoretically, if markets are efficient and prices incorporate all available information, the potential benefits of employee satisfaction lead to higher current valuations and, as a result, there should not be any abnormal future returns for these responsible companies. However, an investor preference for investing in (socially responsible) companies that treat their employees well should push up their stock prices and lower their costs of capital, resulting in lower expected returns to investors in equilibrium (Heinkel et al. 2001).

In sharp contrast, in an influential paper published over a decade ago, Edmans (2011) shows that a portfolio of the "100 Best Companies to Work For in America" (BCs) earns a positive and significant Carhart alpha. Given that this was one of the first papers to show the outperformance of an SRI strategy, it has been heavily cited by academics and practitioners. By now, however, one would expect that the market has priced in employee satisfaction. Moreover, the documented Carhart alpha may simply be the result of exposure to newly discovered factors such as investment, profitability, or quality.

In this paper, we study whether Edmans (2011) result (a) holds out-of-sample, (b) is driven by exposure to newly discovered factors and characteristics, and (c) depends on the state of the economy. To answer these questions, we extend the analysis in the original study in two important dimensions: adding 11 years of more recent data and using several more sophisticated factor models to control for factors discovered since publication. Specifically, we estimate BC portfolio alpha in a sample from April 1984 to December 2020, using the following factors models: the 6-factor model of Fama and French (2018), q4-factor model of Hou et al. (2014), q5-factor model of Hou et al. (2020), and a 6-factor model including betting-against-beta and quality-minus-junk factors (Frazzini and Pedersen 2014; Asness et al. 2019).

We find that up to 22% of the BC portfolio's Carhart alpha could be attributed to exposures to more recently incorporated factors such as investment, profitability, and quality. This result highlights the

[1] Global Sustainable Investment Review, 2020: http://www.gsi-alliance.org/wp-content/uploads/2021/08/GSIR-20201.pdf

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importance of using newer factor models to estimate excess returns. However, we still estimate that the BC equal-weighted portfolio's monthly alpha ranges between a lower bound of 17 basis points (bps) and an upper bound of 23 bps during the period from 1984 to 2020. The alphas are positive and statistically significant across all factor models. We also show that the excess returns are not due to firm characteristics or industry composition. Moreover, the abnormal returns remain virtually unchanged when we exclude small stocks. However, a value-weighted BC portfolio does not generate excess returns that are robust across time and factor models. Our results show that this is because a few mega-cap stocks dominate the value-weighted portfolio. Consistently, we find that abnormal returns are positive, significant, and robust when we use a capped value-weighted portfolio where small BCs have weights according to their size, but no mega-cap stock dominate the portfolio.

Having established positive risk-adjusted returns to the BC portfolio, we investigate whether and how these excess returns vary across periods. In particular, we are interested in whether these excess returns come only from "good" times when the stock market is booming and investor sentiments are high (Pastor et al. 2020; Bansal et al. 2021; Pastor et al. 2021), or, on the contrary, whether the abnormal returns are higher in crisis periods (Lins et al. 2017; Li et al. 2020). The estimated excess returns are positive in most periods during our sample. Importantly, we do not find that BCs outperform *only* in good times. In contrast, the largest excess returns come from bad times, i.e., the crisis periods. In addition, we do not observe any particular (upward or downward) trend in the size of the BC portfolio's alpha during the four-decade sample we study.

Our paper makes two contributions to the literature. Our first contribution comes from extending the analysis in Edmans (2011) in two important dimensions: (a) adding more than one decade of recent data, and (b) using state-of-art factor models and taking into account a more comprehensive set of characteristics to estimate the excess returns. This is an important contribution given the alleged replication crisis in finance (Harvey 2019). For example, in their influential paper, Hou et al. (2018) conclude that "most anomalies fail to hold up to currently acceptable standards for empirical finance."

The extension we are doing is particularly important. First, anomalies tend to disappear after publication (Bebchuk et al. 2013; Mclean and Pontiff 2016) and, if so, previously documented anomalies have no practical relevance anymore. This is even more likely to be the case for employee satisfaction, given the fact that Edmans' work—one of the first to show the outperformance of an SRI strategy—was published over 10 years ago, and has received extensive coverage among practitioners. Moreover, excess returns documented using a Carhart model could simply be due to exposure to more recently recognized factors such as quality and investment. Furthermore, in contrast to Edmans (2011), several recent studies posit and/or provide empirical evidence to show that at least some approaches to SRI must result in or are associated with *lower* returns (Luo and Balvers 2017; Pastor et al. 2020; Zerbib 2020; Barber et al. 2021).

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Our analysis indeed highlights the importance of controlling for factors such as quality and profitability as BCs tend to be companies with higher profitability and quality. However, using the most recent empirical finance models, we find that BCs continue to generate sizable and significant positive excess returns more than one decade after the original publication.

Our second contribution is estimating the BC portfolio excess returns' variability in different periods. Bansal et al. (2021) provide evidence that highly rated ESG stocks underperform in bad times (and on average) but they outperform in good economic conditions due to a "luxury" feature of these assets and more investor demand in "good" periods. Pastor et al. (2020) and Pastor et al. (2021) also argue that while the return to investing in "green" stocks is low in equilibrium, these stocks may outperform in periods where a positive unexpected shock hits the ESG factor due to unexpected shifts in investors' taste for these stocks. Our results, based on a completely different measure of business social responsibility, are not consistent with these ideas. We find that the BC portfolio outperforms in most periods and the highest excess return is in fact associated with the crisis periods. This result is consistent with studies that provide evidence on the importance of corporate culture and corporate social responsibility in bad times such as Lins, Servaes, and Tamayo (2017), Graham et al. (2018), and Li et al. (2020).

Given the significant outperformance of companies with high employee satisfaction, funds could exploit such information. Indeed, there is at least one such fund. The founder of one of the earliest socially responsible funds (in 1984), Jerome Dodson launched Parnassus Workplace Fund (later changed to the Parnassus Endeavor Fund) in 2005; its main investment criteria were workplace quality factors. Indeed, Dodson was inspired to start the fund after seeing promising performance in the initial lists of BCs. Milton Moskowitz, the coauthor of the BC lists, was a consultant to the fund. Dodson managed the fund until December 2020. According to Morningstar, the fund's performance is ranked in the top percentile of all large-cap growth funds over every period from 1-15 years.[2] In an interview in 2013, Dodson explains the reason behind the large difference in the performance of his fund around the financial crisis, compared to the market:

> I think what happens when you have a contented workplace, people are willing to put out more effort to improve operations during really difficult times. While I think every organization has their ups and downs, the downs are not as pronounced because everybody pulls together to try to get through the crisis. And, of course, this consistently more engaged performance inevitably reveals itself in the firm's bottom line.[3]

Still, an important question remains unanswered. More than a decade after the discovery of significant outperformance of an SRI strategy based on employee satisfaction, it is unclear why the outperformance

[2] The 1-year, 3-year, 5-year, 10-year, and 15-year returns are 60.1%, 21.2%, 19.5%, 19%, and 14.6%, respectively. The fund has an annual return of 14% since inception and the highest possible rating for all periods from Morningstar. Data retrieved from Morningstar as of August 31, 2021.

[3] See: https://www.fastcompany.com/3006150/proof-profits-americas-happiest-companies-also-fare-best-financially

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persists. We conclude the paper with offering some potential reasons behind this surprising persistence of outperformance.

2. Data, measurement, and descriptive statistics

2.1. Best Companies List

Our paper is focused on one particular aspect of business social responsibility: the way the firm treats its employees, leading to higher employee satisfaction. The data we use is the list of the "100 Best Companies to Work for in America" (BCs) which is produced by the Great Place to Work Institute. The first list was produced in 1984 and was published in a work of Levering et al. (1984). The list was updated in February 1993 by Levering and Moskowitz (1993). Since 1998, the list is announced each year in *Fortune* around mid-January.

The list is determined by an anonymous survey of employees and is America's largest ongoing annual workforce study, representing more than 4.1 million employees in the most recent year. Companies need to employ at least 1,000 US employees to be considered for inclusion in the list. Importantly, unlike most of ESG scores, BCs are not determined based on analyzing companies' written policies. Measuring workplace culture or employee satisfaction is impossible without knowing how employees feel within the corporation. Consequently, the list is not only considering pay and benefits programs but also employees' trust in their organization and workplace opportunities to reach their full human potential, irrespective of rank and position. These experiences are analyzed relative to each organization's size and the norms prevailing in their industry.[4] More specifically, companies are scored in the following four areas: credibility (communication to employees), respect (opportunities and benefits), fairness (compensation and diversity), and pride/camaraderie (teamwork, philanthropy, and celebrations). The list is the oldest publicly available data source on employee satisfaction, and receives significant attention from the media, shareholders, management, and employees themselves.

Although there are 100 companies in each list, a subset of those will be used in this paper since we require companies to have CRSP stock returns for at least one month during the period of inclusion in the list used for the BC portfolio formation. Table 1 shows the number of companies included in our analyses, per year, from 1984 to 2020. Overall, we have 283 unique public companies in the BC portfolio throughout the sample period, with an average of 50 companies per year. Columns 3-6 of Table 1 report statistics regarding the number of companies that are added and dropped from each list compared to the previous one.

[4] See, https://fortune.com/franchise-list-page/best-companies-2020-methodology/

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While we do observe some level of persistence in the list, which is expected, on average 26% of BCs in a given year are not on the list of the next year. Similarly, on average, more than 20% of firms in a given year list are newly added firms that were not part of previous list. Overall, Table 1 shows that while workplace culture and practices (and hence employee satisfaction) are typically persistent within firms, there is reasonable within-firm variation in employee satisfaction.

To understand the industry composition of companies in the list, Figure 1 plots the total number of BCs throughout the sample categorized by the Fama-French 49-industry classification. As shown, we have BCs in 41 (out of 49) industries, showing that the list includes companies from diverse industries and is not focused on only one or a few of them. However, there is a large cross-sectional variation in the number of BCs per industry. Specifically, BCs tend to be more common in industries that are more dependent on human capital such as Computer Software, Banking, Trading, Pharmaceutical Products, and Electronic Equipment, and much less common in Real Estate, Coal, Utilities, Aircraft, Tobacco Products, and Beer and Liquor. Industries with no publicly traded BC in our sample include Agriculture, Rubber and Plastic Products, Fabricated Products, Shipbuilding and Railroad Equipment, Defense, Precious Metals, Non-Metallic and Industrial Metal Mining. In sum, more than 80% of industries have had at least one BC and there is significant cross-sectional variation in the number of BCs per industry.

We also study the evolution of BCs presence in each industry across years. Figure 2 shows the number of BCs by industry for each year from 1984 to 2020. For easier illustration, the cells are color coded such that the darker the cells are, the larger the numbers are. As shown, prior to 2000, Consumer Goods and Computers (Hardware) had the largest number of BCs. Their importance, however, has been decreasing so that in the last year of the sample we do not have any BCs in these two industries. One industry with a clear upward trend is Computer Software, accommodating about 30% of BCs within itself by the end of the sample. Notable industries with relatively stable presence are Pharmaceutical Products, Restaurants, Hotels, Motels, and Banking.

Table 2 Panel A reports descriptive statistics on important variables and characteristics of BCs in the last year of our data, 2020. In particular, we include variables that are related to the most commonly used factors such as size, value, profitability, and investments. The first important observation is that BCs are large companies, with an average (median) market capitalization of $55bn ($20bn). This shows that BCs tend to be large companies and the size distribution is skewed to the right.

To better understand the size distribution of BCs, both in the cross-section and over time, Figure 3 plots the number of BCs in each year based on their size category. Specifically, we group companies into six size categories according to the NYSE breakpoints: below the 5th percentile, between the 5th and 20th percentiles, between the 20th and 50th, between the 50th and 80th, between the 80th and 95th, and above the 95th percentile. This figure also clearly shows that BCs are typically either large or extremely large

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companies. Importantly, there are very few stocks whose size is below the 20th percentile NYSE breakpoints, and it is rare to observe BCs that are smaller than the NYSE 5th percentile breakpoints (in many years there is no BC in this category). Finally, we do not see any visible change in the distribution of BC size across years. Overall, the main takeaway from these statistics is that BCs are rarely tiny-cap stocks, they are typically large, and a few are extremely large companies.

In Table 2 Panel A, we also see that, on average, BCs have relatively high market-to-book ratios of 12.44 and gross-profit-to-total-assets ratios of 38%, and that BCs spend relatively little on capital expenditures (4% of revenue) and have relatively large amounts of intangibles in their balance sheet (22% of total assets).[5]

2.2. Portfolio construction and return statistics

Our portfolio construction follows Edmans (2011). We form each year's portfolio in the beginning of the month following the BC list announcement. Specifically, since the first list was published in March 1984, we form our portfolio in April 1984. Similarly, since the first update to the list was published in February 1993, we rebalance the original portfolio in March 1993. Starting from 1998, *Fortune* has been announcing the list in mid-January and hence we rebalance our portfolio on the first day of February. It should be noted that by forming the portfolio about two weeks after the announcement, our empirical analysis effectively tests a joint hypothesis that employee satisfaction improves firm value, and that the stock market does not fully consider this public information in the short term.

Our main analysis in this paper is focused on equal-weighted portfolios. The estimated abnormal returns could be sensitive to the choice of equal- or value-weighted portfolio (Fama and French 2008; Hou et al. 2018), given the disproportionate effects that "micro" stocks may have on the returns of equal-weighted portfolios. In our setup, however, this should not be a significant source of concern since, as shown in Figure 3, we have very few small BCs across the years. Nonetheless, in Section 4, we show formally that our results are not driven by small-cap BCs. We also provide estimates of excess returns using value-weighted portfolios (some in Section 4 and some in the Internet Appendix).

We use monthly returns, and our final sample includes 441 months ranging from April 1984 to December 2020. Table 2, Panel B shows descriptive statistics of the BC portfolio return. For the sake of comparison, we have also reported these statistics for two other benchmarks: industry-matched and characteristics-matched portfolios. The industry-matched portfolio is constructed using industry returns

[5] We note that some of these characteristics vary significantly across years. For example, comparing the first available BC list to the one in 2020, we observe that in 1984, operating margin was 1.08% (compared to 20.46% in 2020) and the proportion of intangible assets in the balance sheet was 0.85 (compared to 22.29% in 2020).

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from the Fama-French 49-industry classification, which are matched to the BC portfolio. The characteristics-matched portfolio is formed using the size-value-momentum characteristics matched returns. The average monthly return of the BC portfolio is 1.25%. This is higher than both the returns for the industry-matched (1.10%) and characteristics-matched (0.95%) portfolios. Moreover, the cumulative return of the BC portfolio during the sample is 12,500%, compared with 7,607% for the industry-matched portfolio and 4,039% for the characteristics-matched portfolio. These show significant historical outperformance of BCs compared to their industry peers and to firms with similar size, market-to-book ratios, and recent past performance.

To better illustrate the historical performance of the BC portfolio, Figure 4 plots the performance of a $1,000 initial investment in the BC portfolio compared to those of industry- and characteristics-matched portfolios. For a better visualization, we use two different y-scale for pre- and post-2000, hence two separate graphs. Consistent with the statistics presented above, this figure clearly shows that the BC portfolio has performed significantly better than the benchmarks (i.e., the industry- and characteristics-matched portfolios) from 1984 to 2020.

3. Empirical Methodology and Results

We turn to statistical analysis to evaluate how much (if anything) of the BC portfolio's return could be characterized as abnormal. Specifically, in this section, we start with factor regressions using the most recent empirical asset pricing models to control for factors that are known to be associated with higher returns. We also study whether controlling for characteristics matters for these results using both characteristics-adjusted returns as well as Fama-MacBeth regressions. Finally, we estimate abnormal returns in different periods.

3.1. Time-series factor regressions

The most common practice in empirical finance to estimate abnormal returns attached to a particular strategy/portfolio is to use factor regressions. In these setups, we run a regression of excess returns of the portfolio on known (risk) factors. A general equation for such a regression is:

$$\mathrm{ER_t} = \alpha + \sum \beta_f \cdot \mathrm{Factor_{ft}} + \varepsilon_t \qquad (1)$$

where ER_t is portfolio return in excess of the risk-free rate. *Factor* represents the set of known factors (depending on the choice of model) and β_f is the loading on factor f. α measures abnormal returns that cannot be associated with the exposure to any of the factors.

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When a researcher reports abnormal returns, there is always a critical assumption behind that result: the model of risk-return used to estimate abnormal returns is the true model and does not miss any important risk factor. Indeed, it is common to find studies that report significant alphas that turn out to be due to exposure to risk factors that were not controlled for. As an example, while several studies (Fabozzi et al. 2008; Hong and Kacperczyk 2009; Statman and Glushkov 2009) show that investing in "sin" stocks is associated with significantly positive abnormal returns controlling for up to four factors of the Carhart model, Blitz and Fabozzi (2017) revisits this literature and find that this result is merely due to exposure to two new factors of profitability and investment.

Unfortunately, we can never be sure whether we are using the true model to account for risk. Therefore, it's important to estimate abnormal returns using several most recent alternative models to check the robustness of the results. Specifically, we use the 6-factor model of Fama and French (2018), the q-factor model of Hou et al. (2014), the q5-factor model of Hou et al. (2020), and a 6-factor model following Asness et al. (2019) and Frazzini and Pedersen (2014). We also report the results using the Carhart 4-factor model, used in Edmans (2011), to understand the importance of new factors.

The Fama-French 6-factor model (hereafter FF6) adds profitability (RMW) and investment (CMA) to the Carhart model and has been shown to be significantly better than the previous versions in explaining the cross-section of stock returns (Fama and French 2015; Fama and French 2018). The q-factor model, however, comprises the factors of market (R_MKT), size (R_ME), investment (R_IA), and profitability (R_ROE). The q5-factor model augments the q-model with expected investment growth (R_EG). The q-models have been shown to be able to explain more anomalies than alternative models such as FF6 (Hou et al. 2020). Finally, we also consider a 6-factor model, developed by Asness et al. (2019) and Frazzini and Pedersen (2014), including the market, size, value, momentum (UMD), betting against beta (BAB), and quality minus junk factors (QMJ). We will refer to this model as AFP as per initial letters of the authors' names.

We first start by trying to reproduce the main results of Edmans (2011). Reproducing the exact point estimates of an empirical study is typically notoriously hard unless one uses identical data and codes. Other than various data treatments and options used in estimation that could be done differently, the underlying data used in financial economics frequently change or are updated. This is one reason that makes it challenging to precisely reproduce the results of empirical studies. Important for our study, Akey et al. (2021) show that factor vintages are substantially different, and the difference increases with the length of time between vintages. As a result, they find that keeping everything else similar and only using different factor vintages leads to substantially different regression estimates even for diversified portfolios. They also show that the effects are large across characteristics and therefore portfolios sorted on characteristics (e.g., size, book-to-market, momentum, etc.) are significantly affected. Akey et al. (2021) study 187 high-low

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anomaly portfolios and show that "unconditional alphas of almost a third of portfolios lose statistical significance due to changes in factor vintages." and that these effects are economically large. Therefore, they recommend using different factor vintages for evaluating the robustness of the results of empirical studies.

Using the insights from Akey et al. (2021), in Table 3 we provide the BC excess return for the period 1984 to 2009 and using the Carhart model, as in Edmans (2011), but with different factor vintages.[6] The benchmark used to compute excess returns are either the risk-free rate (Columns 1-3) or the return to a portfolio with similar size, value, and momentum characteristics (Columns 4-6). Within each dependent variable, there are three sets of results. The first column shows the results of the original study, while the second and third columns show our estimates using the November 2011 and May 2021 factor vintages, respectively. For the alpha estimates, we also report Newey-West standard errors with two-lags (in addition to one-lag estimates), which are shown in brackets. It is important to emphasize that the differences between the second and third columns are solely due to differences in factor vintages. The differences between the first and the two subsequent columns could either be due to differences in factor vintages or differences in returns (due to CRSP updates) and/or differences in characteristics (Compustat updates).

The results of Table 3 show that we successfully reproduce the estimated alphas (0.31 and 0.24) in the original study. While using different factor vintages do (slightly) change the point estimates and standard errors, the results are robust across them. In addition, using two-lags to compute the Newey-West standard errors tends to slightly reduce the estimated standard errors but they remain in the same range. While we have tried to follow as closely as possible to the original study, there are some differences in point estimates and standard errors, mostly for size and momentum factor loadings in characteristics-adjusted regressions (Columns 3-6). The only reason that we can think of is changes in the return of portfolios sorted based on characteristics through time (Akey et al. 2021).[7]

Overall, given the challenging nature of replicating empirical studies, we believe our replication is satisfactory. Importantly, these differences do not appear to have any effect on the estimated alphas. We also see very small differences in our alpha's t-statistics with the original study (e.g., 3.48 in Column 3 vs. 3.34 in Column 1, or 3.03 in Column 6 vs. 2.94 in Column 4), but these minor differences are in the same range as differences resulting in merely changing the factor vintages or changing the number of lags used in estimating the standard errors. Therefore, we consider this as a reasonably well replication of the original study.

[6] We downloaded factor vintages from the *wayBackmachine* functionality in www.archive.org. A more complete replication including the industry-adjusted returns and value-weighted BC portfolio is provided in the Internet Appendix Table 1.

[7] The fact that the results of industry-adjusted returns (reported in Internet Appendix) are also very close to the original study is consistent with this idea.

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Having replicated the main result of the original study, we use four more recent asset pricing models (FF6, q-, q5-, and AFP models) as well as the original Carhart 4-factor model to test whether the BC portfolio was and is still associated with abnormal returns. For the sake of comparability, the rest of the paper uses the 2021 factors vintage and reports the results separately for 1984-2009 and 1984-2020.

Table 4 reports the estimates of BC portfolio excess return using different factor regressions for the period 1984-2009 (Columns 1-5) and 1984-2020 (Columns 6-10). There are five columns per sample period, each corresponding to one factor model. The dependent variable is BC portfolio return in excess of the risk-free rate. Our alpha estimates using newer factor models but with the same sample as the original study show that indeed part of the excess returns could be associated with the exposure to newer factors. For example, the excess returns estimated using FF6 (25 bps per month) is about 20% lower than the Carhart alpha as the BC portfolio has a positive loading on profitability factor. Having said that, the estimated alphas stay sizable and statistically significant both in FF6 and the q model. The size of the alpha in the q5 and AFP is still sizable but loses statistical significance (Columns 4 and 5).

When we use the full sample (Columns 6-10), we find positive and significant excess returns for the BC portfolio that are robust across all models. Specifically, the estimated monthly excess return ranges from the lower bound of 17 bps (in the q5 model) to the higher bound of 23 bps (in the Carhart model). The BC portfolio has positive loading on the size factor (ME in q-models and SMB in others) and negative loading on the momentum factor (MOM in FF6 and UMD in AFP). Moreover, it has positive and significant loadings on profitability of FF6 (RMW) and the quality factor of AFP (QMJ). These results suggest that controlling for new factors of profitability and quality are important in this context. Specifically, the magnitude of alpha in FF6 and AFP models that explicitly control for profitability and quality (Columns 7 and 10) are 15% and 22% lower, respectively, than the alpha estimated using the Carhart model.

It is important to highlight that the excess returns we estimate are gross of transaction costs. Given that BCs tend to be very large stocks and the average annual turnover of the portfolio is low (16%), transaction costs are unlikely to be significant (compared to the alpha of between 2 and 2.7% per year we reported above). To be more precise, we follow the logic used in Chaudhuri et al. (2020) to provide an estimated transaction cost. If the mid-point of bid-ask spread reflects the fair value, swapping one stock with another one (one sell and one buy) would incur a transaction cost equivalent to the bid-ask spread. Assuming the average bid-ask spread of 50 bps for a large-cap stock in the U.S. (CORWIN and SCHULTZ 2012) and the average annual portfolio turnover of 16% yields an estimated annual transaction cost of 16% * 50 = 8 bps. Using the most conservative estimated annual alpha of 206 bps annually (i.e., monthly alpha of 17 bps, Column 9 Table 4), we estimate an annual alpha of 198 bps net of transaction costs.

One potential concern related to factor regression results is that differences in characteristics of the BCs (e.g., size, book-to-market ratio) compared to other firms could be the reason behind the abnormal

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returns.[8] To address this issue, we run similar factor regressions as before but this time we use characteristics-adjusted excess returns as the dependent variable. Specifically, instead of using the risk-free rate as the benchmark in computing the BC portfolio excess returns, we deduct the return on a portfolio of stocks with similar size, book-to-market ratio, and momentum. The results of these regressions are shown in Table 5. As shown, the results of these characteristics-adjusted excess returns are in line with our previous findings: although using newer factor models tends to reduce the estimated excess returns, the alphas remain sizable (between 16 and 23 bps per month) and statistically significant regardless of the model (Columns 6-10).

Overall, the results shown in Table 4 and Table 5 have two important messages: (1) Up to about 22% of the BCs' alpha estimated using the Carhart model is attributed to exposures to newer factors such as profitability and quality; and (2) the BC portfolio alpha remains positive and significant in a large sample spanning four decades across several of the most sophisticated factor models.

3.2. Fama-MacBeth regressions

The potential disadvantage of controlling for characteristics as we did in the previous section is that in a factor regression setup, we are bound to use a few characteristics. To be able to control for many more characteristics, we use the Fama and MacBeth (1973) methodology (hereafter FM).[9] Specifically, we start with the universe of publicly traded ordinary common stocks for which we have returns data from CRSP. We then exclude very small firms with market cap below $1 million and stocks with prices lower than $1. Then, we run the following regression:

$$R_{it} = \alpha_0 + \alpha_1 BC_{it} + \alpha_2 Z_{it} + \varepsilon_{it} \qquad (2)$$

where R_{it} is the return on stock *i* in month *t*; BC_{it} is the dummy variable which has the value of one if the stock is among the most recent BC list and zero otherwise. Z_{it} is the set of characteristics variables. α_1, which is the coefficient of interest and measures the abnormal return associated with companies in the BC list.

We use basic characteristics from as in Edmans (2011), and add further characteristics variables that have been used more recently in the literature. For basic characteristics, we try to follow as closely as possible with the definitions mentioned in the original study. Basic characteristics include the log of market

[8] While we do have factors such as size, value, and momentum included as covariates in the factor regressions, these characteristics could still have explanatory power, as argued in Daniel and Titman (1997).
[9] It is important to note that time-series factor models and cross-sectional FM regressions are methodologically different and could yield different results. Fama and French (2019) provide detailed comparison of the two methods.

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cap in month t-2 (*SIZE*), the log of the book-to-market ratio which is recalculated each July (*BM*), the ratio of dividends in the previous fiscal year to the market value at calendar year end (*YIELD*), the logs of the cumulative compounded returns in month t-3 to month t-2 (*RET2-3*), month t-6 to month t-4 (*RET4-6*), month t-12 to month t-7 (*RET7-12*), the log of dollar trading volume in month t-2 (*DVOL*), and the log of price at the end of month t-2 (*PRICE*).

Additional new characteristics we include are asset growth (the annual growth rate of total assets), ROE, cash-based operating profitability, and earning surprise. All these data on additional characteristics are obtained directly from the open source of Chen and Zimmenmann (Forthcoming).

Table 6 reports the results of FM regressions. Similar as before, we report the results both for 1984-2009 (Columns 1-3) and 1984-2020 (Columns 4-6). Within each sample, we have three sets of results. The first column uses the basic characteristics, the second column adds the additional characteristics, and the third column uses industry-adjusted return (instead of raw return) as the dependent variable. It is important to highlight that in addition to factors mentioned earlier that generate differences in our results with those of the original study, our sample selection for FM regressions is different, as discussed before.[10]

Similar as in factor regressions, we find that adding new characteristics reduces the magnitude of the BC excess return in both samples. Specifically, the BC excess return goes down from 57 to 55 bps per month in 1984-2009 and from 49 to 46 bps in 1984-2020. All the new characteristics show up statistically significant in explaining the stock returns. Despite these, we find sizable and statistically significant BC excess returns in both sample and across different specifications. For example, the coefficient of 0.46 in Column 5 suggests that after controlling for all characteristics, BCs earn, on average, an excess return of 46 bps per month during 1984-2020. The corresponding industry-adjusted excess return is also 39 bps per month and statistically significant (Column 6).

As shown in Figure 1, a large fraction of BCs belongs to Computer Software industry (CS). To make sure that our results are not driven solely by companies in this sector, we conducted two further analyses: (a) dropping companies in CS, and (b) using the whole sample but controlling for CS by including a CS dummy variable. The coefficient on BC dummy remains sizable and statistically significant in both analyses (0.35 and 0.43 with t-statistics of 3.30 and 4.43, respectively), showing that our results are not driven solely by IT companies. These results are provided in the Internet Appendix.

Overall, the results of FM regressions show that the excess return to BCs is sizable and that it cannot be associated to the BCs' characteristics or industry composition.

[10] In the Internet Appendix Table 2, we reproduce Table 6 without any of our sampling filters. The results are robust and all the excess return estimates in the table are statistically significant. For example, the corresponding BC coefficient in column 1 is 0.39 (similar as the original study) with the t-statistics of 3.71 and 0.37 in column 4 with the t-statistics of 4.25.

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3.3. BC excess returns in different periods

In the previous sections we documented a significant average excess return for BCs. In this section, we investigate whether abnormal returns associated with BCs vary across different periods. In particular, we are interested in whether these excess returns come only from "good" times when the stock market is booming and investor sentiments are high (Pastor et al. 2020; Bansal et al. 2021; Pastor et al. 2021), or, on the contrary, whether the abnormal returns are higher in crisis periods (Lins et al. 2017; Li et al. 2020).

To estimate alphas in different periods, we add three period dummies (*Boom*, *Crisis*, *Normal*) to our factor models instead of the intercept. We define *Boom* as the periods associated with well-known stock market booms during our sample, namely the dotcom boom from January 1995 to March 2000, and the boom leading to financial crisis from November 2002 to July 2008. *Crisis* is an indicator variable associated with the dotcom crash from April 2000 to October 2002 and the financial crisis from August 2008 to March 2009 (Lins et al. 2017; Li et al. 2020). We put the rest of months together as "normal" periods.

Table 7 provides the estimated excess returns in different periods using various factor models and different benchmarks (the risk-free rate in Panel A and characteristics-matched return in Panel B). There is one clear message from these results: while BCs tend to perform well in most periods, the excess returns are particularly high during bad times. The estimated magnitudes imply that BCs had abnormal returns in the range of 0.8% and 1.2% per-month (depending on the model and specification) during crises periods. These results are not consistent with studies (Pastor et al. 2020; Bansal et al. 2021) that argue that socially responsible companies should underperform in bad times and could only potentially outperform in good times. Instead, our results are in line with several other studies that suggest socially responsible businesses outperform in bad times. For example, Lins, Servaes, and Tamayo (2017) show that during the financial crisis, firms with higher corporate social responsibility had significantly higher stock returns (up to seven percentage points higher) than firms with low social responsibility. Somewhat related to this, executives surveyed in Graham et al. (2018) believe the importance of corporate culture for firm outcomes is more apparent in a challenging operational environment, because a strong culture empowers executives and rank-and-file employees to make consistent decisions and effort based on long-term perspectives. Also, Li et al. (2020) find that strong corporate culture has particularly strong effect on firm performance during bad times.[11]

We also analyze the BC excess return in different periods and across time using FM regressions controlling for the full set of characteristics as in Column 5 of Table 6. An advantage of FM methodology

[11] In the Internet Appendix Table 5, we provide more granular estimates of excess returns for each boom and crisis period. Those results show that the average monthly excess return in the dotcom boom was in the range of 20-30 bps whereas in the boom preceding the financial crisis was around 0. None of the estimated boom excess returns are statistically significant. On the contrary, the alphas for both crisis periods are sizable, statistically significant, and robust across different factor models.

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is that it gives an estimated excess return in each month separately. Using these monthly alphas, we compute the BC cumulative excess return during the sample, which is plotted in Figure 5. As shown, the comulative excess return is going upward in most months, which means that the BC excess return is positive in those periods. Importantly, the comulative excess return continues to climb significantly following the sample period used in Edmans (2011). Finally, we use these monthly excess returns to compute FM alphas in different periods (e.g., Pre- and post-2009, 5- or 10-year intervals, different market conditions). For brevity, these results are provided in the Internet Appendix Table 6. Overall, the results are consistent with the ones reported earlier that the BC excess returns have been positive in most periods and are not coming only from good times. Consistent with the results from factor regressions, when we look at periods with different market conditions, we find that the largest excess return belongs to the financial crisis period (1.4% per month).[12] Finally, there does not seem to be any trend (upward or downward) in the size of abnormal returns over time.

To sum up, we do not find evidence supporting the idea that the BC excess return is merely coming from good times. While the portfolio has performed well in most periods, its largest excess returns come from the crisis periods.

4. Additional Analysis

4.1. Investigating the influence of small-cap stocks

So far, we have shown abnormal returns associated with an equal-weighted BC portfolio. One potential concern with an equal-weighted portfolio is that its returns might be dramatically affected by micro-cap stocks. Although we have shown that the BCs tend to be large or very large companies, we do have some micro-cap BCs (see Figure 3). To be more precise, Table 8 provides the number of BCs, grouped by NYSE size categories for each year in our sample. As shown, on average, we have about two BCs per-year whose size is between the NYSE 5th and 20th percentile breakpoints (i.e., micro-cap stocks) and less than one company per-year with the size below the NYSE 5th percentile breakpoint (i.e., nano-cap stocks).

To investigate the influence of very small stocks on our reported excess returns, we form the BC portfolio excluding (a) only nano-cap BCs (size < NYSE 5th percentile) or (b) both nano-cap and micro-cap stocks (size < NYSE 20th percentile). We then estimate the excess returns to these portfolios using all the factor models used in the paper. These results, together with our original estimates for comparison, are

[12] One notable difference between FM and factor model results is that while still positive, the FM alpha in the dotcom crash is much smaller than the one estimated from factor regressions.

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provided in Panel A of Table 9. As shown, the estimated alphas barely change in either their magnitude or statistical significance when we exclude very small BCs.

We also construct value-weighted (VW) portfolios as an alternative analysis to lessen the influence of micro-cap stocks on the BC portfolio return. While constructing a VW portfolio limits the influence of micro-cap stocks, one or a few stocks might dominate the portfolio and make the portfolio unbalanced. This is indeed the case in our setup: the largest three BCs typically constitute about 40-50% of the VW portfolio. To illustrate this, Figure 6 plots market capitalizations of the first and third largest BCs compared to the NYSE 95th percentile breakpoint for each year during our sample. As shown, the top three largest BCs are extremely large companies even when compared to the NYSE 95th percentile breakpoint. More precisely, for example in 2020, the first, second, and third largest BCs have market capitalizations of $332bn, $244bn, and $230bn, compared with NYSE 80th and 95th percentile breakpoints of $17bn and $76bn, respectively.

In the presence of extremely large companies in the portfolio, to construct a tradable but balanced portfolio, we use capped VW portfolios in which the market caps are winsorized at a threshold such as the NYSE 80th percentile (Jensen et al. 2021). This effectively means that we assume that the BCs with market capitalizations larger than the NYSE 80th percentile breakpoint have market caps equal to the NYSE 80th percentile. This way, smaller companies will have small weights in the portfolio but also no one stock will dominate the portfolio. We use both the straight and capped VW BC portfolios and estimate their excess returns using different factor regressions.

Panel B of Table 9 reports the excess returns across all models for straight VW (rows 1 and 2) and capped VW (rows 3 and 4) portfolios.[13] Rows 1 and 3 report the results using 1984-2009 sample, whereas rows 2 and 4 show the estimates for 1984-2020 sample. The results of rows 1 and 2 show that the excess return to the straight VW BC portfolio is not robust across different factor models and time periods. Specifically, while the point estimates and their statistical significance go up in FF6 and q model (compared to the original Carhart estimate of 0.29), the q5-alpha and that of the AFP shrink in size and become statistically insignificant. When we use the full sample in row 2, again we find results that are not robust across models; even the Carhart alpha drops significantly to 8 bps per month. However, from the results of Panel A, we already know that this lack of robustness in VW (compared to EW) results cannot be due to the large influence of tiny-cap stocks. Instead, it is most likely due to mega-cap BCs that dominate the VW portfolio. Indeed, the results in rows 3 and 4 show a sizable and robust excess return to the capped VW BC portfolio (between 14 and 22 bps per month during 1984-2020). The results of characteristics-adjusted VW portfolios, provided in Internet Appendix Tables 7-10, are also consistent with the ones reported here.

[13] For the sake of brevity, we only report alphas here. We provide detailed estimates in the Internet Appendix Tables 7 and 8. We also provide similar tables but using characteristics-adjusted excess return as the dependent variable for both the straight and capped value-weighted portfolios (Appendix Tables 9 and 10).

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Overall, the results of Table 9 highlight that the BC portfolio excess return is not driven by small-cap stocks and that a capped VW BC portfolio also earns significant abnormal returns. The large difference in the estimated alphas between the straight and capped VW portfolios highlights the following point: unlike in usual situations, when dealing with a portfolio of mostly large and few mega-cap companies, the results of an EW or capped VW portfolio may be more informative than a straight VW portfolio, which could be dominated by few stocks.

4.2. Small and Medium BCs

As shown in previous sections, BCs tend to be very large companies. One might wonder whether employee satisfaction is also undervalued in small and medium enterprises (SMEs). We expect that the undervaluation should be larger, if anything, in SMEs. Specifically, if the stock market undervalues employee satisfaction in a sample of large well-known companies with huge media coverage, such an undervaluation should be higher in smaller, less-known companies. Moreover, one would expect an even higher undervaluation when employee satisfaction data are not publicly available or not announced publicly with large media coverage. Therefore, we expect that the excess return to employee satisfaction documented so far is a lower bound estimate.

In this section, we attempt to provide some preliminary results on the relation between employee satisfaction and stock returns in SMEs. Perhaps the best source of the data, somewhat comparable to what we have used in the paper, for SMEs comes from Glassdoor.com website. Based on employees' comments and reviews of companies, Glassdoor announces the list of best SMEs to work for (SME BCs). These data unfortunately do not go back far in time and only available since 2015. This gives us 6 years of data from January 2015 to December 2020. The announcement dates in this period have been in December of the previous year. For example, the 2015 list is announced in December 2014.

Each list is composed of 50 SMEs. However, as expected, most of companies in the list are not publicly traded (or we do not have their return data from CRSP) and therefore could not be used in our analysis. As a result, our portfolio of great SMEs to work for in 2015 has 10 stocks. These include, for example, Chegg Inc, Opower Inc., Arista Networks Inc, and Castlight Health Inc. We rebalance our portfolio in the first day of January each year by incorporating new SMEs that are included in the most current list. In total, there are 29 unique SMEs included in our portfolio.

We first analyze the historical performance of the SME BC portfolio from 2015 to 2020. The portfolio has an average annual return of 33.7%, compared with 10.5% for S&P 500 and 10.1% for Russell 2000. In addition, the cumulative portfolio return in the period was 471% versus 82% and 78% for S&P 500 and Russell 2000 respectively. These clearly show that the SME BC portfolio has a great performance during the sample.

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We also analyze the risk-adjusted performance of the portfolio using different factor models used in the paper. The results are shown in Table 10. As shown, these excess returns are much larger than what we reported in Table 4 for (large) BCs. For example, the Fama-French 6-facor monthly alpha for the SME BC portfolio is 0.78, which is significantly more than the corresponding number (0.20) in Table 4. This is consistent with the idea that the excess returns to employee satisfaction should be larger in smaller companies that receive less attention and coverage. Having said that, we acknowledge again that due to data limitations, these results should only be taken as suggestive.

5. Concluding remarks

In this paper we use a sample spanning four decades from 1984 to 2020 and the state-of-art empirical asset pricing models to study the relationship between employee satisfaction and long-run stock returns. We find that an equal-weighted portfolio of companies with the highest employee satisfaction earns an abnormal return of about 20 bps per month in our 36-year sample. We also show that these positive excess returns are present in most periods during the sample, do not follow any detectable upward or downward trend, and are highest during the crisis periods.

Edmans (2011) concludes with two distinct scenarios going forward: (1) If the market learns of the significant impact of employee satisfaction on future returns, the excess returns should disappear; or (2) "If the market does not update (e.g., because intangibles are inherently difficult to incorporate into stock prices) and arbitrage remains limited, the superior returns may persist going forward." Our results appear to be more consistent with the second scenario. An interesting research question in 10 years would be whether BCs still outperform.

Still, an important question remains unanswered. More than a decade after the discovery of significant outperformance of an SRI strategy based on employee satisfaction and despite the huge rise of SRI, it is unclear why the outperformance persists. One potential reason could be that the focus of most funds and investors have been on exclusionary/negative screening rather positive or best-in-class screening. Consistently, in their survey of professional investment managers, Amel-Zadeh and Serafeim (2018) find that more than 80% of investors do consider ESG information when making investment decisions. Among all ESG investment styles they look at, the most popular style among large asset managers is negative screening whereas positive/best-in-class screening is the least popular style.

Another potential reason behind the persistent outperformance of BCs could be that responsible funds have been more focused on the "E" (environment) and "G" (governance) dimensions of social responsibility rather than on the "S" (social) aspect. And within the S dimension, investors may have been targeting factors such as pay ratio and (gender) diversity rather than employee wellbeing. Furthermore, most approaches to

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ESG investing are predominantly based on ESG scores and easily quantifiable factors. These approaches ignore important qualitative factors such as fairness, respect, pride, and camaraderie that are used to measure employee satisfaction.

Our work has a broader message for portfolio managers or investors: The value of obtaining and/or constructing reliable data on employee satisfaction for a larger set of companies could be significant. More specifically, if the stock market undervalues employee satisfaction information that is publicly announced and receives huge coverage, such an undervaluation could be larger for companies that receive less attention. This, however, would require reliable employee satisfaction data on a larger set of companies. Obtaining such data is surely feasible in practice. For instance, Glassdoor.com website includes millions of employees' comments -working for thousands of different companies- about various aspects of their workplace. Such textual data could be processed by investors to construct portfolios based on particular aspects of workplace culture the investor is interested in. Indeed, there has been efforts by various companies to assess corporate culture and employee satisfaction within firms using textual analysis as well as surveying of employees. One such enterprise is the investment research firm Irrational Capital that provides proprietary data/scores on the level of employee engagement, motivation, and satisfaction within firms. These scores are generated by analyzing public and proprietary data, mostly coming from surveys of employees including those from Glassdoor website. Using these data, Harbor Capital Advisors has launched in February 2022 an ETF called Harbor Corporate Culture Leaders (ticker: HAPY).[14] Overall, the main insight from our paper, i.e., that firms with high level of employee satisfaction outperform, can and should be applied to a much broader sample of firms to better exploit potential undervaluation of employee satisfaction by the stock market.

Our work has several limitations. First, our focus is on employee satisfaction, however, we encourage future research to address the impact of other aspects of business social responsibility including protecting the environment, addressing racism, accounting for consumers' health, encouraging gender diversity, reducing excessive pay gaps, respecting human rights, and confronting child labor. Second, our data include only companies with the highest employee satisfaction and do not identify those that perform badly. Due to potentially asymmetric relations, whether companies with low levels of employee satisfaction earn positive, negative, or no abnormal returns remains an open empirical question.

Third, while the particular aspect of social responsibility (i.e., employee satisfaction) that we study here pays off for shareholders in our four-decade sample, there are surely other factors that require a trade-off between social responsibility and financial performance. Identifying those factors would be important for shareholders to form realistic expectations. It would also be useful for policymakers to design better

[14] As we do not have access to these data, we cannot judge their reliability and value-added. The point is to provide an example that asset managers can and do use proprietary information about employee satisfaction and corporate culture to form portfolios including these criteria.

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incentives and policies to improve social outcomes when it is too costly for shareholders to invest. Finally, we only investigated the effect of employee satisfaction on returns to shareholders. Studying its impact on other stakeholders, such as creditors and consumers, would be another fruitful research area.

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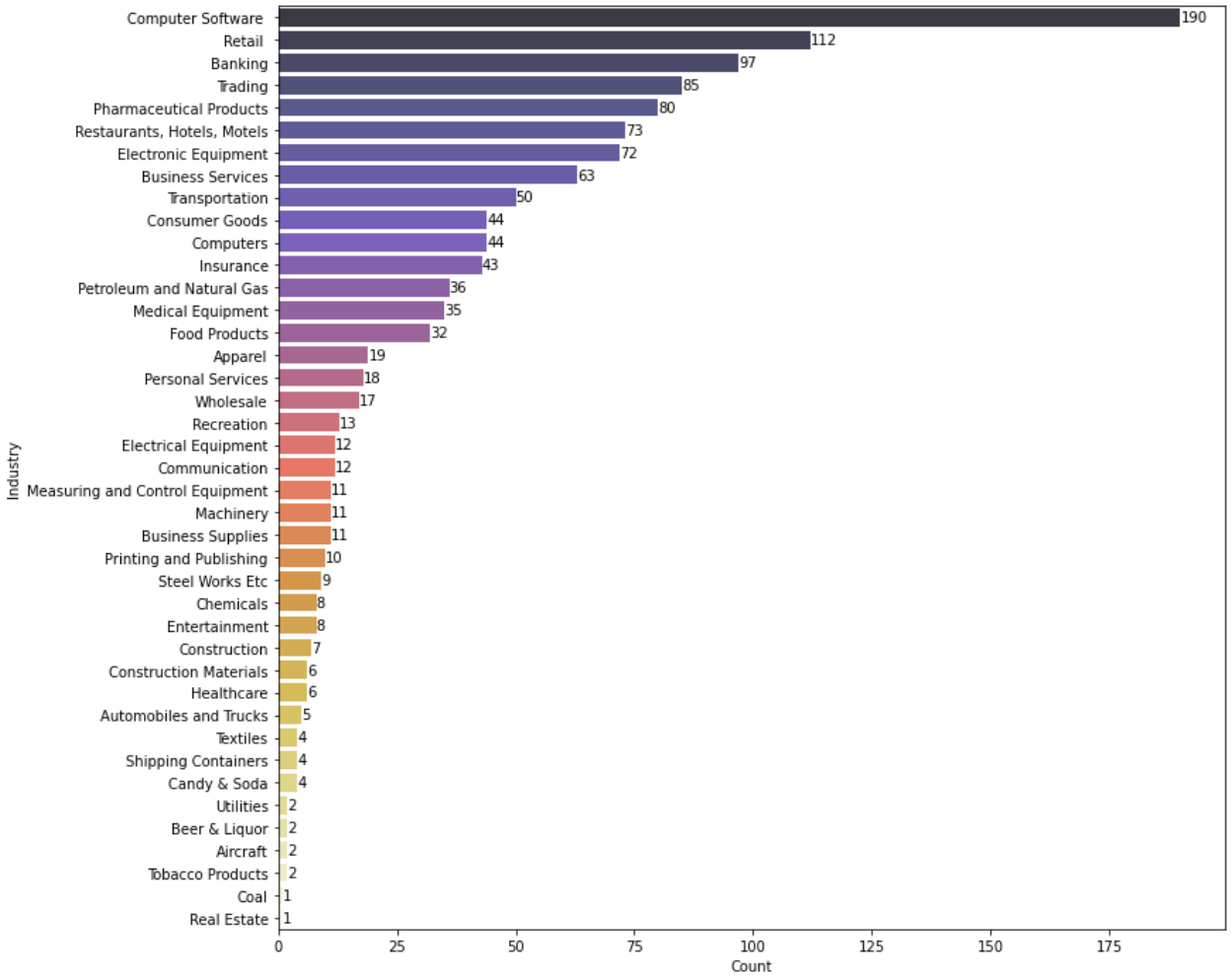
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Figure 1. Counts of BCs, by industry

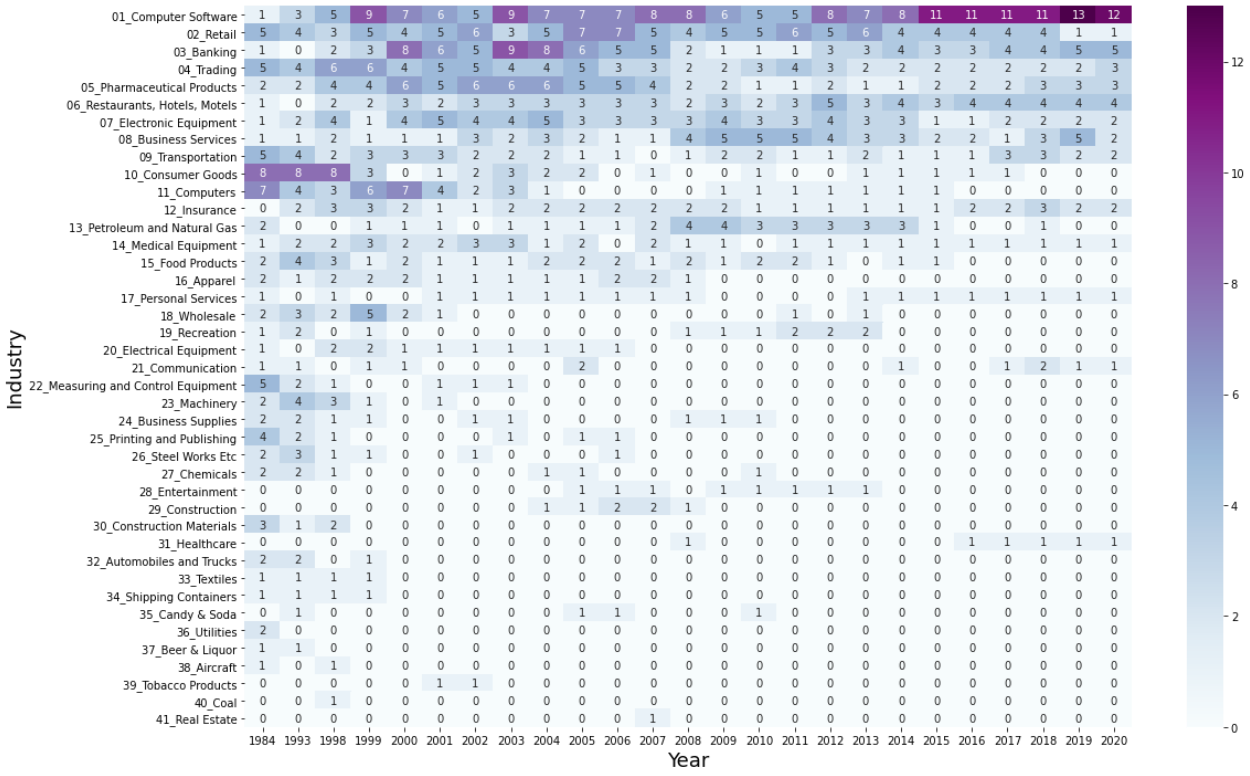
This graph shows the number of BCs during the whole sample from 1984-2020 categorized by industry (Fama-French classifications).



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Figure 2. Evolution of BCs' industry composition

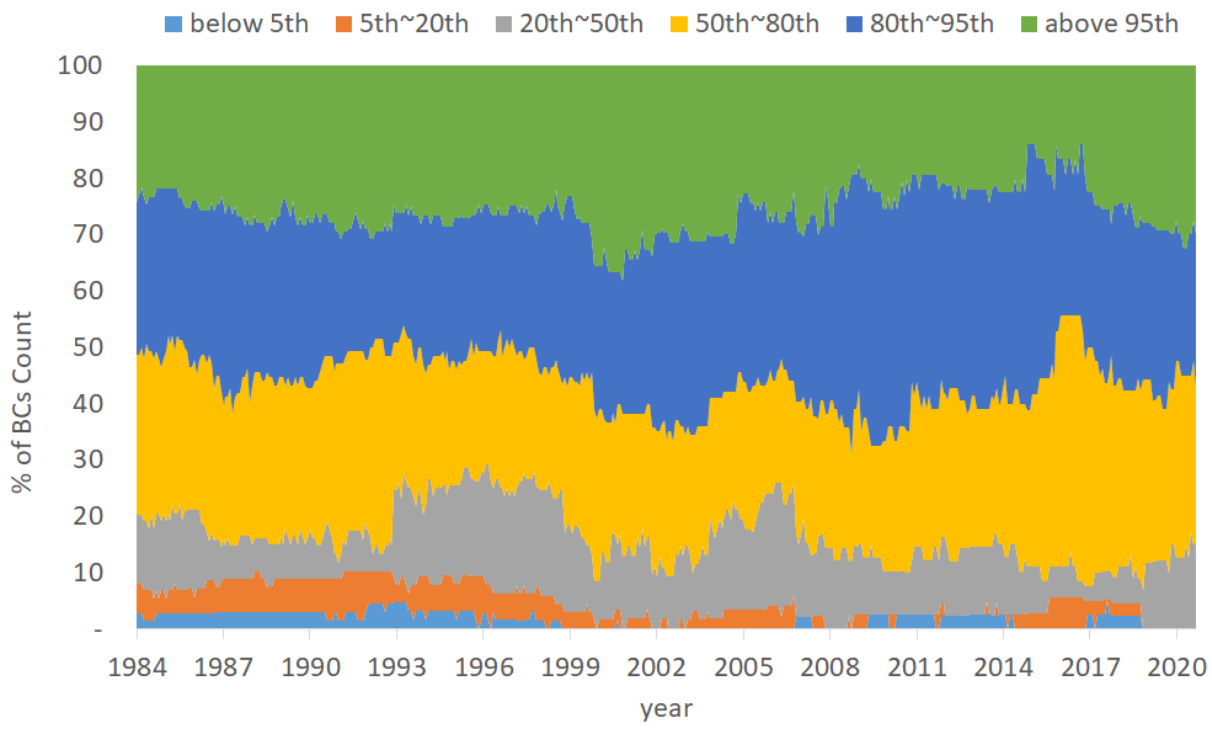
This figure shows the number of BCs by industry and year. The darker the cell color is, the larger the number is.



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Figure 3. Distribution of BCs according to NYSE size categories across time

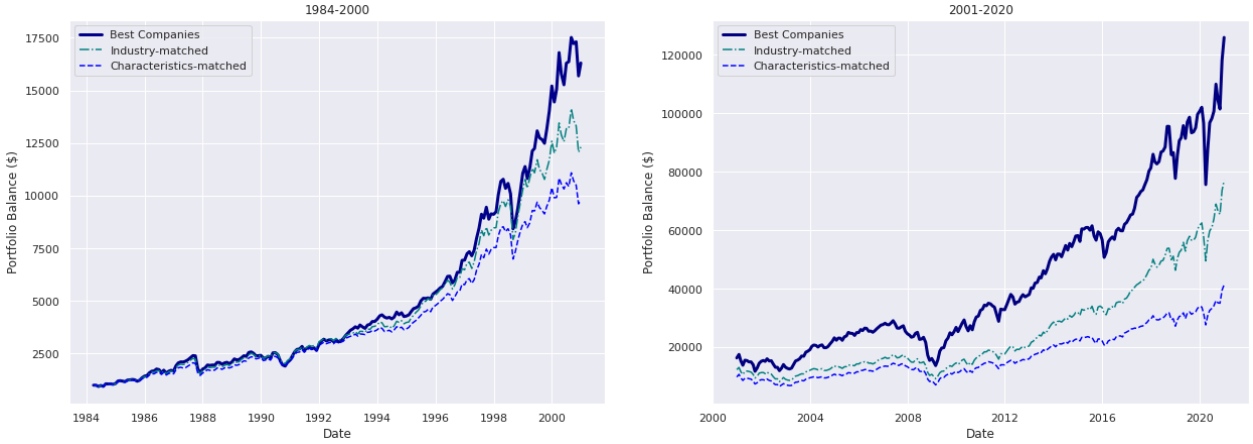
This figure plots the percentage of BCs that belong to each NYSE size category for each year from 1984-2020. The categories are below NYSE 5th percentile breakpoint (nano-caps), between 5th and 20th percentiles (micro-caps), between 20th and 50th percentiles (small caps), between 50th and 80th percentiles (large caps), between 80th and 95th percentiles (mega caps) and above 95th percentile ("giga" caps). NYSE different percentile breakpoints are taken from Ken French website.



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Figure 4. BC portfolio performance against benchmarks

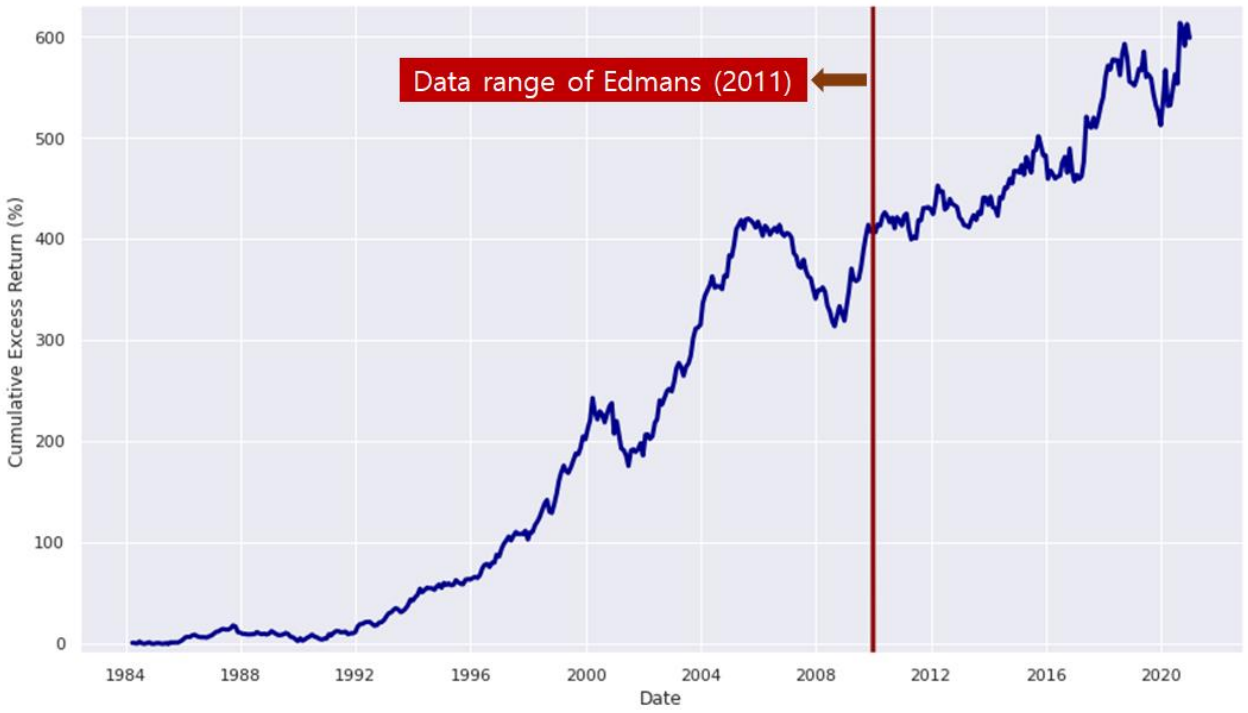
This figure plots the performance of $1000 initial investment in BCs' equal-weighted portfolio along with the performance of two benchmarks: Characteristics-matched (based on size, value and momentum) and industry-matched portfolios. The sample includes 441 months ranging from April 1984 to December 2020. For a better visualization, we use two different y-scales for pre- and post-2000, hence two separate graphs.



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Figure 5. BC portfolio cumulative excess returns using FM regressions

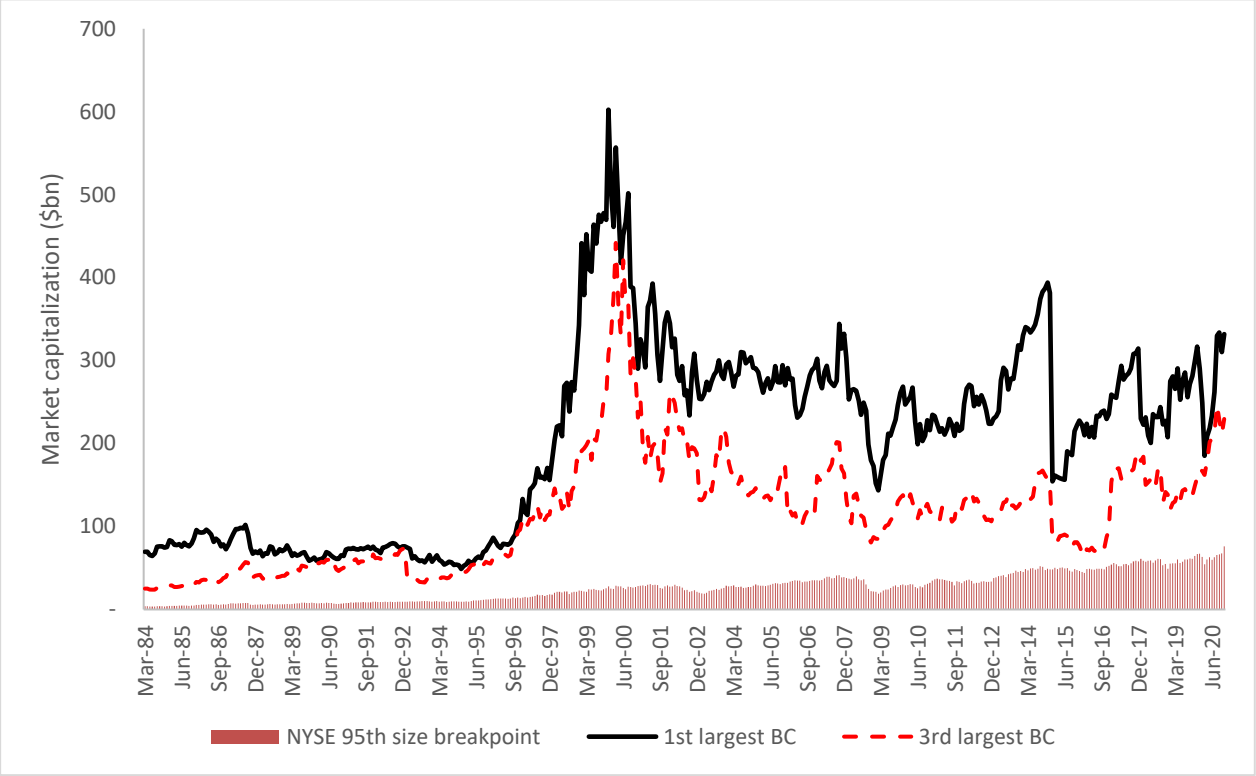
This plot shows the cumulative excess return of the BC portfolio during the sample, from April 1984 to December 2020. The model used to estimate the monthly excess returns is Fama-MacBeth regressions with the full set of characteristics as in Table 6 column 5.



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Figure 6. Largest BCs compared to NYSE 95th percentile breakpoint across time

This plot shows the market capitalization of the largest (solid line) and the third largest (dashed line) BCs throughout our sample period. The bars represent NYSE 95th percentile breakpoint in each year for the sake of comparison.



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Table 1. Summary statistics on number of BCs used in the paper

This table reports (in Column 2) the number of Best Companies per year that are included in the BC portfolio, i.e., those that had returns available on CRSP for at least one month during the portfolio formation periods of each year. Column 3 shows the number of companies that are added to the list and Column 4 reports the number of added companies as a fraction of total number of companies in the list. Columns 5 and 6 repeat the similar statistics (the raw number and the proportion) for companies that are dropped from the list in each year.

Year of list	# of Best Companies	# Added	(%)	# Dropped	(%)
1984	78				
1993	69	30	(43.5)	39	(50.0)
1998	70	35	(50.0)	34	(49.3)
1999	68	27	(39.7)	29	(41.4)
2000	60	20	(33.3)	28	(41.2)
2001	55	15	(27.3)	20	(33.3)
2002	55	13	(23.6)	13	(23.6)
2003	61	14	(23.0)	8	(14.5)
2004	57	11	(19.3)	15	(24.6)
2005	58	11	(19.0)	10	(17.5)
2006	50	8	(16.0)	16	(27.6)
2007	47	10	(21.3)	13	(26.0)
2008	43	12	(27.9)	16	(34.0)
2009	42	9	(21.4)	10	(23.3)
2010	40	8	(20.0)	10	(23.8)
2011	41	7	(17.1)	6	(15.0)
2012	45	7	(15.6)	3	(7.3)
2013	42	5	(11.9)	8	(17.8)
2014	40	6	(15.0)	8	(19.0)
2015	36	6	(16.7)	10	(25.0)
2016	36	4	(11.1)	4	(11.1)
2017	40	6	(15.0)	2	(5.6)
2018	45	7	(15.6)	2	(5.0)
2019	43	7	(16.3)	9	(20.0)
2020	40	3	(7.5)	6	(14.0)
Average	50	12	(23)	13	(26)

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Table 2. Descriptive statistics of BCs in 2020

Panel A of this table shows summary characteristics for the most recent year (2020) of Best Companies in the paper. The list was published in *Fortune* that was public on January 2020. The statistics are based on the whole firm-month observations for the period of February to December 2020.
Panel B of the table shows summary statistics for the BC equal-weighted portfolio return together with the same statistics for industry- and characteristics-matched portfolios for the sake of comparison. The industry-matched portfolio is constructed by the matched industry returns from the Fama-French 49 industries classification with the same weights of each firm in the BC portfolio. The characteristics-matched portfolio is constructed using the method of Daniel et al. (1997) and Wermers (2003) and having the same weights for firms in the BC portfolio. The sample includes 441 months ranging from April 1984 to December 2020.

Panel A: Descriptive Statistics on BCs' Characteristics, 2020

	# firms	Mean	Median	Std.	Min	Max
Market Cap ($ bn)	40	55.02	19.55	72.25	0.78	333.93
Market/book	33	12.44	4.98	22.00	0.69	112.30
Dividend yield (%)	34	1.26	0.59	1.57	0.00	8.26
Gross profits to total assets (%)	32	38.15	34.97	21.56	9.58	92.89
Capex to revenue (%)	40	4.14	3.12	3.83	0.00	16.13
Operating cash flows to revenue (%)	40	20.46	19.17	16.66	-2.99	66.03
Intangibles as a % of total assets (%)	40	22.29	16.13	21.90	0.00	77.04

Panel B: Descriptive statistics of BC portfolio return and benchmarks, 1984-2020

	Mean	Std.	25th	Median	75th	Cumulative return (%)
Best Companies	1.25	5.05	-1.57	1.41	4.35	12,500
Industry-matched	1.10	4.74	-1.58	1.43	3.99	7,607
Characteristics-matched	0.95	4.44	-1.46	1.29	3.47	4,039

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Table 3. Replicating the original study with alternative factor vintages and standard error computations

This table provides estimates of the excess return of the equal-weighted BC portfolio. The setup is similar as in Edmans (2011) in that the model used is the Carhart four-factor model and the sample is from April 1984 to December 2009. In Columns 1-3, the excess return is computed using the risk-free as the benchmark whereas in Columns 4-6 the benchmark is the return to a portfolio with similar size, value, and momentum characteristics. Columns 1 and 4 report the corresponding results in Edmans (2011). The only difference between Columns 2 and 3 (also Columns 5 and 6) is the choice of the factor vintage. Columns 2 and 5 use the 2011 factor vintage whereas columns 3 and 6 uses the 2021 factor vintage. Newey-West standard errors with one-lag are computed and the associated t-statistics are reported in parentheses. For alpha estimates, we also report t-statistics based on two-lags, shown in brackets. **, and *** indicate significance levels at 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Dep var:	Excess returns over					
Benchmark:	Risk-free			Characteristics		
Factor vintage:	2010	2011	2021	2010	2011	2021
	Edmans (2011)			Edmans (2011)		
α	0.31***	0.32***	0.31***	0.24***	0.24***	0.24***
	(3.34)	(3.45)	(3.48)	(2.94)	(3.06)	(3.03)
		[3.35]	[3.37]		[2.98]	[2.95]
Mkt	1.08***	1.08***	1.09***	0.09***	0.12***	0.13***
	(41.01)	(41.97)	(44.07)	(3.69)	(6.50)	(6.79)
HML	0.03	0.02	0.05	0.01	-0.00	0.01
	(0.70)	(0.48)	(1.11)	(0.45)	(-0.14)	(0.30)
SMB	0.17***	0.17***	0.19***	0.05	0.12***	0.12***
	(3.66)	(3.61)	(4.17)	(1.39)	(4.47)	(4.67)
MOM	-0.15***	-0.15***	-0.15***	-0.09***	-0.00	-0.00
	(-6.36)	(-6.37)	(-6.59)	(-4.80)	(-0.10)	(-0.07)
Observations	309	309	309	309	309	309

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Table 4. Excess returns of the BC portfolio

This table provides estimates of the excess return of the equal-weighted BC portfolio using different sample periods as well as alternative factor models. The excess return is computed using the risk-free as the benchmark. The factor models used are: Carhart 4-factor, Fama-French 5-factor plus momentum (FF6), q, q-5, and AFP (Asness, Frazzini, and Pedersen) 6-factor model. The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). The q-model has market, size (R_ME), investment (R_IA) and profitability (R_ROE), and the q5 models adds to these the expected investment growth factor (R_EG). The AFP model includes the market, size, value and momentum (UMD), and adds to them betting against beta (BAB) and quality minus junk (QMJ) factors. The sample starts from April 1984 and ends in either December 2009 (Columns 1-5) or December 2020 (Columns 6-10). Newey-West standard errors with one-lag are computed and the t-statistics are reported in parentheses. **, and *** indicate significance levels at 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Sample:	1984-2009					1984-2020				
Model:	Carhart	FF6	q	q5	AFP	Carhart	FF6	q	q5	AFP
α	0.31***	0.25***	0.27**	0.21	0.13	0.23***	0.20***	0.21**	0.17**	0.18**
	(3.48)	(2.71)	(2.52)	(1.84)	(1.40)	(3.00)	(2.60)	(2.53)	(2.02)	(2.29)
Mkt	1.09***	1.10***	1.10***	1.11***	1.18***	1.08***	1.08***	1.08***	1.09***	1.12***
	(44.07)	(44.17)	(36.29)	(35.83)	(36.97)	(49.85)	(52.98)	(41.71)	(41.56)	(41.82)
SMB	0.19***	0.24***			0.35***	0.18***	0.22***			0.26***
	(4.17)	(6.20)			(8.24)	(4.09)	(6.21)			(6.46)
HML	0.05	-0.02			0.16***	0.01	-0.01			0.02
	(1.11)	(-0.47)			(3.11)	(0.29)	(-0.17)			(0.41)
MOM	-0.15***	-0.16***				-0.13***	-0.13***			
	(-6.59)	(-7.22)				(-5.83)	(-6.13)			
RMW		0.13***					0.13***			
		(3.33)					(3.35)			
CMA		0.01					-0.09			
		(0.17)					(-1.52)			
R_ME			0.13**	0.14**				0.13***	0.14***	
			(2.33)	(2.33)				(2.64)	(2.65)	
R_IA			0.00	-0.02				-0.07	-0.07	
			(0.06)	(-0.22)				(-1.19)	(-1.30)	
R_ROE			-0.11	-0.14**				-0.11**	-0.13**	
			(-1.81)	(-2.05)				(-2.06)	(-2.16)	
R_EG				0.09					0.07	
				(1.24)					(0.98)	
UMD					-0.14***					-0.12***
					(-6.08)					(-5.48)
BAB					-0.11***					-0.04
					(-3.71)					(-1.40)
QMJ					0.36***					0.19***
					(6.04)					(3.34)
# obs	309	309	309	309	309	441	441	441	441	441

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Table 5. Characteristics-adjusted excess return of the BC portfolio

This table provides estimates of the excess return of the equal-weighted BC portfolio using different sample periods as well as alternative factor models. The excess return is the BC portfolio return in excess of the return on a portfolio with similar size, value and momentum characteristics. The factor models used are: Carhart 4-factor, Fama-French 5-factor plus momentum (FF6), q, q-5, and AFP (Asness, Frazzini, and Pedersen) 6-factor model. The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). The q-model has market, size (R_ME), investment (R_IA), and profitability (R_ROE), and the q5 models adds to these the expected investment growth factor (R_EG). The AFP model includes the market, size, value, and momentum (UMD), and adds to them betting against beta (BAB) and quality minus junk (QMJ) factors. The sample starts from April 1984 and ends in either December 2009 (Columns 1-5) or December 2020 (Columns 6-10). Newey-West standard errors with one-lag are computed and the t-statistics are reported in parentheses. **, and *** indicate significance levels at 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Sample:	1984-2009					1984-2020				
Model:	Carhart	FF6	q	q5	AFP	Carhart	FF6	q	q5	AFP
α	0.24***	0.20**	0.23***	0.16	0.17	0.18***	0.16**	0.21***	0.16**	0.17**
	(3.03)	(2.48)	(2.71)	(1.65)	(1.95)	(2.62)	(2.31)	(2.89)	(2.13)	(2.21)
Mkt	0.13***	0.14***	0.12***	0.13***	0.17***	0.15***	0.15***	0.14***	0.14***	0.16***
	(6.79)	(6.80)	(6.40)	(6.68)	(6.41)	(7.25)	(7.73)	(7.21)	(7.80)	(6.39)
SMB	0.12***	0.13***			0.20***	0.11***	0.13***			0.16***
	(4.67)	(4.87)			(6.22)	(4.03)	(4.38)			(4.53)
HML	0.01	-0.05			0.09	0.00	-0.02			0.02
	(0.30)	(-1.34)			(1.91)	(0.04)	(-0.37)			(0.43)
MOM	-0.00	-0.01				0.01	0.00			
	(-0.07)	(-0.25)				(0.29)	(0.24)			
RMW		0.04					0.06			
		(1.30)					(1.79)			
CMA		0.06					-0.03			
		(1.17)					(-0.44)			
R_ME			0.11***	0.12***				0.10***	0.11***	
			(4.63)	(4.69)				(4.00)	(4.42)	
R_IA			-0.01	-0.03				-0.07	-0.07	
			(-0.21)	(-0.67)				(-1.48)	(-1.62)	
R_ROE			-0.01	-0.04				-0.01	-0.04	
			(-0.19)	(-1.02)				(-0.32)	(-0.99)	
R_EG				0.11					0.07	
				(1.90)					(1.16)	
UMD					0.02					0.02
					(0.74)					(0.85)
BAB					-0.09***					-0.03
					(-3.19)					(-1.33)
QMJ					0.15***					0.07
					(3.04)					(1.43)
# obs	309	309	309	309	309	441	441	441	441	441

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Table 6. Excess returns of the BC portfolio: Fama-MacBeth regressions

This table reports the results of FM regressions for the period 1984-2009 (Columns 1-3) and 1984-2020 (Columns 4-6). Within each time period, there are three specifications: The difference between the first and second specification is the number of characteristics used. The last specification uses industry-adjusted return (instead of raw return) as the dependent variable. *BC* is an indicator variable if the company is in the most recent "Best Companies to Work For" list. *Size* is the log of market cap in month t-2, *BM* is the log of book-to-market ratio, *YIELD* is the ratio of dividends in the previous fiscal year to the market value at calendar year-end, *RETn1-n2* (for different n1 and n2) are the logs of cumulative compounded returns between t-n2 and t-n1. *DVOL* is the log of dollar trading volume in month t-2, *PRC* is the log price in month t-2, *Asset Growth* is the annual rate of total assets, *ROE* is the return on equity, *Cash based op. prof.* is the cash-based operating profitability, and earnings surprise is the earning-per-share (EPS) minus EPS twelve month ago minus drift scaled by the standard deviation of that expression. Drift is the average earnings growth over the past two years. The data on new characteristics (those added in the second specification) are obtained directly from Chen and Zimmenmann (Forthcoming). The sample includes only ordinary common stocks (CRSP share codes of 10 and 11) are included and excludes stocks with the stock price below $1 or the market cap below $1 million. Newey-West standard errors are calculated, and the *t*-statistics are reported in parentheses. **, and *** indicate significance levels at 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	1984-2009			1984-2020		
Dep var:	Raw return		Ind-adj return	Raw return		Ind-adj return
BC	0.57***	0.55***	0.54***	0.49***	0.46***	0.39***
	(5.34)	(4.56)	(5.08)	(5.57)	(4.69)	(4.47)
Size	0.20**	0.18	0.18**	0.18**	0.18**	0.18***
	(2.21)	(1.76)	(2.32)	(2.51)	(2.26)	(2.83)
BM	0.22***	0.20**	0.20***	0.11	0.07	0.09
	(2.82)	(2.35)	(3.04)	(1.67)	(1.03)	(1.62)
YIELD	-0.00	-0.00	-0.00	-0.01**	-0.00	-0.01**
	(-1.09)	(-0.56)	(-0.92)	(-2.08)	(-1.42)	(-2.00)
RET2-3	0.36	0.27	0.19	0.07	0.01	-0.08
	(1.11)	(0.84)	(0.68)	(0.26)	(0.03)	(-0.34)
RET4-6	0.34	0.23	0.25	-0.02	-0.07	-0.07
	(1.24)	(0.80)	(1.01)	(-0.07)	(-0.30)	(-0.35)
RET7-12	0.45**	0.34	0.23	0.17	0.11	0.01
	(2.29)	(1.71)	(1.40)	(1.01)	(0.69)	(0.04)
DVOL	-0.13	-0.09	-0.10	-0.13	-0.11	-0.11
	(-1.57)	(-1.05)	(-1.29)	(-1.93)	(-1.57)	(-1.74)
Price	-0.91***	-1.04***	-1.01***	-0.78***	-0.88***	-0.86***
	(-6.03)	(-6.50)	(-6.46)	(-7.00)	(-7.45)	(-7.44)
Asset growth		0.31***	0.31***		0.30***	0.29***
		(5.76)	(6.13)		(6.85)	(6.76)
ROE		-0.05***	-0.05***		-0.03***	-0.04***
		(-2.81)	(-2.92)		(-2.59)	(-2.75)
Cash based op. prof.		1.64***	1.43***		1.31***	1.11***
		(7.52)	(6.43)		(7.07)	(5.82)
Earnings surprise		0.05***	0.05***		0.05***	0.05***
		(5.95)	(6.04)		(6.40)	(6.44)
Constant	2.62***	2.67***	1.67***	2.53***	2.58***	1.43***
	(3.65)	(3.49)	(2.87)	(4.50)	(4.38)	(3.17)
Observations	1,197,148	891,307	891,286	1,598,644	1,191,344	1,191,318
R-squared	0.05	0.05	0.04	0.04	0.05	0.04
Number of months	309	309	309	441	441	441

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Table 7. BC excess returns in different periods using time series factor regressions

This table provides estimates of the excess return to the equal-weighted BC portfolio in different periods. The excess returns are estimated using time series factor regressions and replacing the constant with three dummy variables representing different periods. *Boom* is an indicator variable associated with the dotcom boom (from January 1995 to March 2000) and the boom leading to financial crisis (from November 2002 to July 2008). *Crisis* represents the dotcom crash (from April 2000 to October 2002) and the financial crisis (from August 2008 to March 2009). *Normal* is the rest of months in the sample. We report alphas using five different factor models including Carhart four-factor, Fama-French 5-factor plus momentum (FF6), q, q-5, and AFP (Asness, Frazzini, and Pedersen) 6-factor model. The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). The q-model has market, size (R_ME), investment (R_IA), and profitability (R_ROE), and the q5 models adds to these the expected investment growth factor (R_EG). The AFP model includes the market, size, value, and momentum (UMD), and adds to them betting against beta (BAB) and quality minus junk (QMJ) factors. Panel A uses BC portfolio return over the risk-free rate whereas Panel B uses the characteristics-adjusted excess return as the dependent variable in factor regressions. The sample is from April 1984 to December 2020. Newey-West standard errors are calculated, and the *t*-statistics are reported in parentheses. **, and *** indicate significance levels at 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)
Model:	Carhart	FF6	q	q5	AFP
Panel A: Excess returns over risk-free					
Normal	0.16	0.16	0.20	0.18	0.17
	(1.72)	(1.71)	(1.80)	(1.60)	(1.71)
Boom	0.14	0.15	0.05	0.04	0.07
	(0.53)	(0.61)	(0.50)	(0.34)	(0.64)
Crisis	1.08***	0.93***	1.16***	1.13***	0.84***
	(3.37)	(2.83)	(3.14)	(3.06)	(2.66)
Panel B: Characteristics-adjusted excess returns					
Normal	0.08	0.08	0.13	0.10	0.09
	(0.89)	(0.88)	(1.44)	(1.03)	(0.94)
Boom	0.10	0.11	0.12	0.10	0.09
	(0.35)	(0.35)	(0.05)	(0.01)	(0.33)
Crisis	0.90***	0.83***	0.90***	0.86***	0.85***
	(3.25)	(2.86)	(3.03)	(2.86)	(2.98)
Observations	441	441	441	441	441

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Table 8. Number of BCs in NYSE size categories, by year

Year End	below 5th	5th~20th	20th~50th	50th~80th	80th~95th	above 95th	Total
1984	2	2	11	20	22	16	73
1985	2	3	10	20	18	18	71
1986	2	3	6	20	21	17	69
1987	2	4	5	19	18	19	67
1988	2	4	4	19	19	19	67
1989	2	4	5	19	18	19	67
1990	1	5	5	22	16	19	68
1991	1	6	5	22	15	20	69
1992	3	4	4	22	16	19	68
1993	2	4	10	16	15	17	64
1994	2	4	9	16	14	18	63
1995	1	5	11	16	15	17	65
1996	1	3	11	17	15	17	64
1997	2	2	14	14	15	17	64
1998	0	3	14	13	20	19	69
1999	0	2	7	22	16	21	68
2000	0	2	8	14	14	22	60
2001	0	1	8	13	15	18	55
2002	0	0	7	13	17	17	54
2003	0	2	6	14	21	18	61
2004	0	2	10	12	16	17	57
2005	0	2	12	11	19	14	58
2006	0	3	10	9	17	11	50
2007	0	1	6	10	13	12	42
2008	0	0	5	8	21	8	42
2009	1	0	4	8	18	9	40
2010	1	0	3	10	17	9	40
2011	0	1	4	11	16	9	41
2012	1	0	5	10	17	9	42
2013	1	1	5	11	15	9	42
2014	0	1	4	11	15	9	40
2015	0	2	2	12	12	8	36
2016	0	2	1	16	12	5	36
2017	1	1	2	15	9	11	39
2018	1	1	2	15	14	12	45
2019	0	0	4	14	11	12	41
2020	0	0	6	11	11	12	40
Average	0.8	2.2	6.6	14.7	16.0	14.7	55.1

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Table 9. Sensitivity of excess returns of the BC portfolio to micro- and mega-cap stocks

Panel A of this table reports the sensitivity of the excess return of an equal-weighted BC portfolio when excluding nano-cap or nano- and micro-cap BCs. Nano caps are defined as stocks with a market cap below the NYSE 5th percentile breakpoint while micro caps are those with market cap between the 5th and 20th percentiles. Panel B reports excess returns of value-weighted BC portfolio with and without capping the weights at 80th percentile of NYSE size breakpoints. It shows value-weighted alphas for both 1984-2009 and 1984-2020. We report alphas using five different factor models including Carhart four-factor, Fama-French 5-factor plus momentum (FF6), q, q-5, and AFP (Asness, Frazzini, and Pedersen) 6-factor model. The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). The q-model has market, size (R_ME), investment (R_IA), and profitability (R_ROE), and the q5 models adds to these the expected investment growth factor (R_EG). The AFP model includes the market, size, value, and momentum (UMD), and adds to them betting against beta (BAB) and quality minus junk (QMJ) factors. Newey-West standard errors are calculated, and the *t*-statistics are reported in parentheses. The sample is from April 1984 to December 2020. **, and *** indicate significance levels at 5%, and 1%, respectively.

Dependent Variable:	Excess return over risk-free				
Model:	Carhart	FF6	q4	q5	AFP
Panel A: Equal-weighted					
α , Full sample	0.23***	0.20***	0.22**	0.17**	0.18**
	(3.00)	(2.60)	(2.57)	(2.06)	(2.24)
α excluding nano caps (size < NYSE 5th pct breakpoint)	0.22***	0.19**	0.21**	0.16**	0.17**
	(2.88)	(2.53)	(2.54)	(1.96)	(2.19)
α excluding nano and micro caps (size < NYSE 20th pct breakpoint)	0.23***	0.20***	0.22***	0.16**	0.19**
	(3.20)	(2.89)	(2.77)	(2.07)	(2.47)
Panel B: Value-weighted (straight and capped)					
α (1984-2009)	0.29**	0.34***	0.41***	0.10	0.17
	(2.34)	(2.98)	(3.30)	(0.81)	(1.49)
α (1984-2020)	0.08	0.17	0.21**	-0.01	0.10
	(0.90)	(1.85)	(2.14)	(-0.06)	(1.05)
α (1984-2009), weights capped	0.30***	0.27***	0.31***	0.19	0.16
	(3.69)	(3.08)	(3.40)	(1.90)	(1.79)
α (1984-2020), weights capped	0.22***	0.22***	0.24***	0.14**	0.21***
	(3.09)	(3.16)	(3.34)	(1.99)	(2.88)

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Table 10. Excess returns of small and medium BCs

This table reports the excess return of an equal-weighted portfolio including publicly traded companies in the "Small & Medium Great Places to Work For" list, announced by Glassdoor.com website. The portfolio is formed/rebalanced in the first day of the month following the list announcement. We add new companies in the list to the portfolio but keep any company that at least once appears in the list. The dependent variable is the portfolio return in excess of the risk-free rate. The factor models used are: Carhart four-factor, Fama-French 5-factor plus momentum (FF6), q, q-5, and AFP (Asness, Frazzini, and Pedersen) 6-factor model. The factors in FF6 are market (Mkt), size (SMB), value (HML), momentum (MOM), profitability (RMW), and investment (CMA). The q-model has market, size (R_ME), investment (R_IA) and profitability (R_ROE), and the q5 models adds to these the expected investment growth factor (R_EG). The AFP model includes the market, size, value and momentum (UMD), and adds to them betting against beta (BAB) and quality minus junk (QMJ) factors. The sample is from January 2015 to December 2020. Newey-West standard errors with one-lag are computed and the t-statistics are reported in parentheses. **, and *** indicate significance levels at 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)
Factor model:	Carhart	FF6	q	q5	AFP
α	0.64	0.78	1.17**	0.81	0.90
	(1.30)	(1.67)	(2.31)	(1.68)	(1.76)
Mkt	1.48***	1.40***	1.05***	1.14***	1.34***
	(11.61)	(8.18)	(6.78)	(7.30)	(8.16)
SMB	1.30***	1.02***			1.28***
	-6.46	(4.79)			(4.35)
HML	-0.58***	-0.50**			-0.97***
	(-2.84)	(-2.65)			(-3.81)
MOM	0.31	0.23			
	(1.83)	(1.30)			
RMW		-0.86**			
		(-2.61)			
CMA		-0.83**			
		(-2.03)			
R_ME			0.78***	0.915***	
			(3.35)	(3.628)	
R_IA			-1.63***	-1.361***	
			(-4.64)	(-4.035)	
R_ROE			-0.57***	-0.936***	
			(-2.73)	(-3.203)	
R_EG				0.71	
				(1.77)	
UMD					0.31
					(1.20)
BAB					-0.23
					(-0.83)
QMJ					-0.53
					(-1.95)
Observations	72	72	72	72	72

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